

1. What is the difference between a local and a global variable?

A:

- **Local Variable:** Declared and used within a function or procedure. Its scope is limited to that block.
 - **Global Variable:** Declared outside of all functions, accessible anywhere in the program.
 - **Example:**
 - DECLARE @local_var INT -- Local variable
 - SELECT @@GLOBAL_VARIABLE_NAME -- Global variable in SQL Server
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2. What is the difference between a Local temporary table and a Global temporary table? How is each one used?

A:

- **Local Temporary Table:** Prefix with #. Visible only to the session that created it.
 - **Global Temporary Table:** Prefix with ##. Visible to all sessions until explicitly dropped or the session ends.
 - **Example:**
 - CREATE TABLE #LocalTemp (ID INT)
 - CREATE TABLE ##GlobalTemp (ID INT)
-

3. What are cursors? Name four types of cursors and when each one would be applied?

A:

- **Cursors:** Allow row-by-row processing of query results.
 - **Types:**
 - **Static Cursor:** Used for stable snapshots of data.
 - **Dynamic Cursor:** Reflects changes in the underlying data.
 - **Forward-Only Cursor:** Iterates rows in one direction.
 - **Keyset-Driven Cursor:** Tracks changes to key values.
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4. What is the purpose of UPDATE STATISTICS?

A:

- **UPDATE STATISTICS:** Refreshes metadata about data distribution in tables or indexed views.

- Purpose: Optimizes query plans by providing up-to-date information to the query optimizer.
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5. How do you load large data to the SQL Server database?

A:

- Use **bulk insert** commands or tools like bcp.
 - Use **SSIS** (SQL Server Integration Services) for ETL processes.
 - Disable indexes and constraints before load for better performance.
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6. How do you check the performance of a query and how do you optimize it?

A:

- Use **SQL Server Profiler** and **Execution Plans**.
 - Optimize with:
 - Indexing strategies.
 - Query refactoring (avoid nested subqueries).
 - Use of SET STATISTICS commands.
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7. What is SQL Server Agent?

A:

- A background service that automates jobs like backups, indexing, and other scheduled tasks in SQL Server.
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8. What is indexing? Advantages and disadvantages of indexes?

A:

- **Indexing:** Creates a data structure for faster retrieval.
 - **Advantages:** Improves query performance.
 - **Disadvantages:** Slows write operations and uses extra storage.
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9. What is de-normalization and when do you do it?

A:

- **De-normalization:** Combines tables to reduce joins, improving query performance in read-heavy systems.

10. Can we rewrite subqueries into simple SELECT statements or with joins? Example?

A:

- Yes, subqueries can often be rewritten with joins.

- **Example:**

-- Using subquery

```
SELECT Name FROM Employees WHERE ID IN (SELECT EmpID FROM Sales);
```

-- Rewritten with JOIN

```
SELECT Employees.Name
```

```
FROM Employees
```

```
JOIN Sales ON Employees.ID = Sales.EmpID;
```

11. What is a function? Give some examples.

A:

- A reusable block of code that returns a value.

Example:

```
CREATE FUNCTION GetFullName(@FirstName NVARCHAR(50), @LastName  
NVARCHAR(50))
```

```
RETURNS NVARCHAR(100)
```

```
AS
```

```
BEGIN
```

```
RETURN @FirstName + ' ' + @LastName
```

```
END
```

12. What is a stored procedure?

A:

- A precompiled SQL code block that performs actions like data manipulation or querying.
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13. Difference between Function and Procedure-in general?

A:

- **Function:** Returns a value and is used in SELECT statements.

- **Procedure:** Performs actions but doesn't necessarily return a value.
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14. Can a stored procedure call another stored procedure? If yes, what level and can it be controlled?

A:

- Yes, one procedure can call another.
 - Nesting is limited to **32 levels**, configurable with recursion settings.
-

15. Can a stored procedure call itself (recursive)? If yes, what level and can it be controlled?

A:

- Yes, procedures can call themselves recursively.
 - **Maximum recursion level:** 32 by default, configurable with MAXRECURSION.
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16. How do you find the number of rows in a table?

A:

SELECT COUNT(*) FROM TableName;

17. What is a table called, if it does not have neither Cluster nor Non-cluster Index?

A:

- Such a table is called a **Heap Table**.
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18. Explain basic SQL queries with SELECT, FROM, WHERE, ORDER BY, GROUP BY, HAVING.

A:

SELECT: Retrieves columns.

FROM: Specifies table.

WHERE: Filters rows.

GROUP BY: Aggregates rows.

HAVING: Filters groups.

ORDER BY: Sorts results.

19. Explain the new features of SQL Server 2008 R2.

A:

- **PowerPivot for Excel** integration.
 - **Master Data Services** for data consistency.
 - **StreamInsight** for real-time analytics.
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20. Explain sp_configure commands, set commands?

A:

- **sp_configure**: Configures server-level settings.
 - **SET Commands**: Modify session-level settings, e.g., SET NOCOUNT ON.
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21. What is the basic function of master, msdb, tempdb databases?

A:

- **master**: Manages system-level info.
 - **msdb**: Manages SQL Agent jobs.
 - **tempdb**: Temporary object storage.
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22. What is a job?

A:

- A scheduled task managed by SQL Server Agent.
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23. What are primary keys and foreign keys?

A:

- **Primary Key**: Unique identifier for a record.
 - **Foreign Key**: References a primary key in another table.
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24. How would you update the rows divisible by 10, given a set of numbers in a column?

A:

UPDATE TableName

SET ColumnName = NewValue

WHERE ColumnName % 10 = 0;

25. How are M-M relationships implemented?

A:

- With a **junction table** containing foreign keys referencing both tables.
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26. How do you know which index a table is using?

A:

```
SELECT * FROM sys.indexes WHERE object_id = OBJECT_ID('TableName');
```

27. How will you test a stored procedure taking two parameters?

A:

```
EXEC ProcedureName @FirstName = 'John', @LastName = 'Doe';
```

28. How do you find errors and the number of rows affected by the last SQL statement?

A:

- Use @@ERROR and @@ROWCOUNT.
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29. How can you get @@error and @@rowcount at the same time?

A:

```
BEGIN TRY
    -- Query
END TRY
BEGIN CATCH
    SELECT @@ERROR AS ErrorCode, @@ROWCOUNT AS RowCount;
END CATCH
```

30. What are sub-queries? In which case are they not feasible?

A:

- A query within another query.
 - Not feasible when performance is critical; better to use joins.
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31. What are the types of joins? When do we use Outer and Self joins?

A:

- **Inner Join:** Common rows in both tables.
- **Outer Join:** Includes unmatched rows (Left/Right/Full).
- **Self Join:** A table joined with itself for hierarchical data.

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